Qualification Pack

Junior Chip Designer (A-Level 'Chip Design')

QP Code: NIE/ELE/Q0102

Version: 1.0

NSQF Level: 5

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NIE/ELE/Q0102: Junior Chip Designer (A-Level 'Chip Design')

Brief Job Description

This course focuses on VLSI technology, covering CMOS transistor theory, logic design, and physical design from RTL to GDSII. It provides in-depth training on Verilog RTL coding, static timing analysis, and FPGA programming. Students will learn to design, analyze, and optimize VLSI circuits using industry-standard tools while gaining hands-on experience in verification techniques such as UVM and System Verilog. The course also addresses physical verification, testability design, and methodologies for ensuring robust chip functionality.

Personal Attributes

A Chip Design Engineer should possess strong analytical and problem-solving skills to address complex circuit design and optimization challenges. Attention to detail is essential for ensuring accuracy in RTL coding, timing analysis, and circuit verification. The ability to adapt to new tools and technologies in the fast-evolving semiconductor industry is crucial. Effective communication and teamwork skills are also important, as the role often involves collaboration with cross-functional teams on VLSI design projects

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. NIE/ELE/N0101: VLSI Fundamentals
2. NIE/ELE/N0102: Verilog RTL coding for Synthesis
3. NIE/ELE/N0103: Static Timing Analysis of VLSI Circuits
4. NIE/ELE/N0104: FPGA Architecture and Programming
5. NIE/ELE/N0105: VLSI Verification fundamentals
6. NIE/ELE/N0106: ASIC Verification using System Verilog and UVM
7. NIE/ELE/N0107: VLSI Circuits Design for testability
8. NIE/ELE/N0108: VLSI Physical Design and Verification
9. NIE/ELE/N0109: Accelerator design using HLS programming
10. NIE/ELE/N0110: SOC Design and Verification
11. NIE/ELE/N0111: Implementation of Advanced VLSI Circuit Design and Verification using Verilog, FPGA, and System-on-Chip (SoC) Architectures
12. DGT/VSQ/N0103: Employability Skills (90 Hours)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	VLSI Design
Country	India
NSQF Level	5
Credits	26
Aligned to NCO/ISCO/ISIC Code	5
Minimum Educational Qualification & Experience	Completed 3 year UG degree (Completed UG Diploma in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience OR Completed 3 year diploma after 10th (Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 1.5 years of experience OR Completed 2nd year diploma after 12th (Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 4.5 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 4 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 3 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA

Next Review Date	30/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-04-EH-02593-2024-V1-NIELIT
NQR Version	1

NIE/ELE/N0101: VLSI Fundamentals

Description

VLSI Fundamentals provides a comprehensive overview of essential concepts in VLSI design, starting with an introduction to VLSI design principles and CMOS transistor theory. Participants will gain insights into CMOS inverter characteristics, logic design, and the intricacies of transistor-level schematics, layouts, and on-chip wire modeling. The course further delves into gate delay modeling, logical effort analysis, critical path optimization techniques, and timing analysis for sequential circuits. By the end of this course, participants will have a solid understanding of key fundamentals in VLSI design necessary for developing efficient and high-performance integrated circuits.

Scope

The scope covers the following :

- The scope of the course on VLSI Fundamentals is to provide a comprehensive understanding of the essential principles and practices in VLSI design. It covers various aspects crucial for designing integrated circuits, starting with an introduction to VLSI design principles and an in-depth exploration of CMOS transistor theory. Participants will learn about CMOS inverter characteristics and logic design, gaining insights into how to design basic logic gates and complex logic functions using CMOS technology.

Elements and Performance Criteria

Introduction to VLSI Design & CMOS Transistor Theory

To be competent, the user/individual on the job must be able to:

PC1. Describe the evolution and significance of VLSI technology, highlighting its impact on modern electronics and computing

PC2. Outline and differentiate between front-end and back-end VLSI design processes, and define MOSFET, explaining its operation principles and modes (cut-off, triode, saturation) along with their significance.

CMOS Inverter Characteristics and Logic Design

To be competent, the user/individual on the job must be able to:

PC3. Describe the functionality and design of CMOS implementations for basic logic gates (AND, OR, NAND, NOR), and design and analyze combinational logic circuits using CMOS gates, focusing on performance metrics.

PC4. Explain factors influencing inverter sizing and apply optimization techniques for performance and power consumption improvements, and combine basic CMOS gates to implement and optimize complex logic functions for speed or area efficiency.

Transistor Schematic, Layouts & On-chip wire modeling

To be competent, the user/individual on the job must be able to:

PC5. Develop transistor-level schematics for basic circuits and apply layout techniques to ensure manufacturability and performance, evaluating the impact of layout on circuit performance.

PC6. Explain design rules for different process nodes, optimize metal layer usage and interconnect routing for performance, and apply design rules and guidelines for efficient layout.

Gate Delay, Logical Effort, Critical Path Optimization and Timing analysis for Sequential circuits

To be competent, the user/individual on the job must be able to:

PC7. Calculate gate delays based on MOSFET parameters, predict gate delays at different process nodes and operating conditions, and optimize logic gate designs for a balance between delay and power consumption using P/N ratio analysis.

PC8. Define logical effort and its relevance in circuit design, apply logical effort principles to estimate delay and power consumption, identify critical paths in combinational logic circuits, and apply gate sizing and logic restructuring techniques to optimize critical paths.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understanding the significance of VLSI technology and its role in modern electronics.

KU2. Analyzing noise margin and power consumption considerations in CMOS circuits.

KU3. Understanding schematic design, layout considerations, and design rules for integrated circuits.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Participants will hone their ability to analyze complex problems and devise solutions in VLSI design by understanding the significance of VLSI technology and the design flow. They will learn to apply critical thinking skills to evaluate noise margin, power consumption, and optimization techniques in CMOS inverter characteristics, logic design, and critical path optimization.

GS2. Learners will gain proficiency in using design tools and techniques essential for VLSI design, such as schematic design, layout considerations, and simulation of on-chip wires. They will develop skills in transistor-level design techniques, gate delay modeling, and logical effort analysis, enabling them to effectively navigate the complexities of CMOS transistor theory and logic design.

GS3. Participants will enhance their communication and collaboration skills. They will learn to effectively communicate their ideas, findings, and design solutions to peers and stakeholders. Collaboration on tasks such as bonding diagram design and packaging considerations will foster teamwork and the ability to work effectively in multidisciplinary settings within the semiconductor industry.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to VLSI Design & CMOS Transistor Theory	**25**	**20**	**-**	**5**
PC1. Describe the evolution and significance of VLSI technology, highlighting its impact on modern electronics and computing	-	-	-	-
PC2. Outline and differentiate between front-end and back-end VLSI design processes, and define MOSFET, explaining its operation principles and modes (cut-off, triode, saturation) along with their significance.	-	-	-	-
CMOS Inverter Characteristics and Logic Design	**25**	**20**	**-**	**5**
PC3. Describe the functionality and design of CMOS implementations for basic logic gates (AND, OR, NAND, NOR), and design and analyze combinational logic circuits using CMOS gates, focusing on performance metrics.	-	-	-	-
PC4. Explain factors influencing inverter sizing and apply optimization techniques for performance and power consumption improvements, and combine basic CMOS gates to implement and optimize complex logic functions for speed or area efficiency.	-	-	-	-
Transistor Schematic, Layouts & On-chip wire modeling	**25**	**20**	**-**	**5**
PC5. Develop transistor-level schematics for basic circuits and apply layout techniques to ensure manufacturability and performance, evaluating the impact of layout on circuit performance.	-	-	-	-
PC6. Explain design rules for different process nodes, optimize metal layer usage and interconnect routing for performance, and apply design rules and guidelines for efficient layout.	-	-	-	-
Gate Delay, Logical Effort, Critical Path Optimization and Timing analysis for Sequential circuits	**25**	**20**	**-**	**5**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC7. Calculate gate delays based on MOSFET parameters, predict gate delays at different process nodes and operating conditions, and optimize logic gate designs for a balance between delay and power consumption using P/N ratio analysis.	-	-	-	-
PC8. Define logical effort and its relevance in circuit design, apply logical effort principles to estimate delay and power consumption, identify critical paths in combinational logic circuits, and apply gate sizing and logic restructuring techniques to optimize critical paths.	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0101
NOS Name	VLSI Fundamentals
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0102: Verilog RTL coding for Synthesis

Description

The VLSI Design syllabus provides an introduction to VLSI technology, its applications, and the design flow. It covers the fundamentals of Register Transfer Level (RTL) design, including the use of Hardware Description Languages (HDLs) like Verilog or VHDL. Students will learn digital logic design principles, focusing on combinational and sequential logic, Boolean algebra, logic gates, flip-flops, registers, and state machines. Practical skills are developed through RTL design using HDL, simulation, and debugging, with a focus on functional verification techniques. Additionally, the course addresses timing constraints and analysis, teaching optimization techniques and resolving setup and hold time violations.

Scope

The scope covers the following :

- Gain a comprehensive understanding of VLSI technology, its significance, and applications, providing a solid foundation for further learning and understand the complete VLSI design flow, including the basics of Register Transfer Level (RTL) design, its process, and methodology.

Elements and Performance Criteria

RTL Design Methodology

To be competent, the user/individual on the job must be able to:

PC1. Understand the basics of Register Transfer Level (RTL) design.

PC2. Demonstrate proficiency in the RTL design process and methodology

Digital Logic Design Principles

To be competent, the user/individual on the job must be able to:

PC3. Apply combinational and sequential logic design concepts effectively.

PC4. Utilize Boolean algebra and logic gates to design digital circuits.

RTL Design Using HDL

To be competent, the user/individual on the job must be able to:

PC5. Demonstrate proficiency in Verilog syntax and constructs.

PC6. Design combinational and sequential logic circuits using HDL.

RTL Simulation and Verification

To be competent, the user/individual on the job must be able to:

PC7. Develop and utilize test benches for comprehensive testing.

PC8. Perform simulation and debugging of RTL designs to ensure functionality and correctness.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Grasping the essence of VLSI technology and its versatile applications across industries. Acquiring knowledge about the complete VLSI design flow, from conceptualization to fabrication, and its significance.

KU2. Developing a solid understanding of digital logic principles, including combinational and sequential logic.

KU3. Acquiring skills in functional verification techniques, test bench development, and timing analysis essential for robust RTL designs.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Developing problem-solving skills and analytical thinking to tackle complex RTL design challenges effectively

GS2. Enhancing communication skills for effectively conveying design ideas and collaborating with team members on RTL design projects

GS3. Cultivating attention to detail to ensure accuracy in RTL design, simulation, and verification, while also adapting to changing design requirements and methodologies

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
RTL Design Methodology	**25**	**20**	**-**	**5**
PC1. Understand the basics of Register Transfer Level (RTL) design.	-	-	-	-
PC2. Demonstrate proficiency in the RTL design process and methodology	-	-	-	-
Digital Logic Design Principles	**25**	**20**	**-**	**5**
PC3. Apply combinational and sequential logic design concepts effectively.	-	-	-	-
PC4. Utilize Boolean algebra and logic gates to design digital circuits.	-	-	-	-
RTL Design Using HDL	**25**	**20**	**-**	**5**
PC5. Demonstrate proficiency in Verilog syntax and constructs.	-	-	-	-
PC6. Design combinational and sequential logic circuits using HDL.	-	-	-	-
RTL Simulation and Verification	**25**	**20**	**-**	**5**
PC7. Develop and utilize test benches for comprehensive testing.	-	-	-	-
PC8. Perform simulation and debugging of RTL designs to ensure functionality and correctness.	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0102
NOS Name	Verilog RTL coding for Synthesis
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0103: Static Timing Analysis of VLSI Circuits

Description

This course provides a comprehensive overview of Static Timing Analysis (STA) in VLSI design, covering timing analysis fundamentals, including races, hazards, and setup and hold timing in both combinational and sequential circuits. Practical examples of setup and hold time violations and solutions are explored, alongside timing constraints for synthesis and circuit synthesis techniques. Advanced topics include timing performance improvement strategies such as pipelining and retiming, clock skew management, and open and closed-loop timing methodologies, with a focus on both block and chip level timing analysis. Students gain hands-on experience in Static Timing Analysis (STA) using Electronic Design Automation (EDA) tools, including timing analysis post-synthesis and place-and-route, interpretation of timing reports, Engineering Change Order (ECO) flows, and timing closure to meet sign-off requirements.

Scope

The scope covers the following :

- Gain insight into timing analysis principles for combinational and sequential circuits, including races, hazards, and setup/hold timing, with practical examples illustrating setup/hold time violations and their resolutions.
- Acquire knowledge of timing constraints essential for synthesis and circuit synthesis techniques, establishing a strong foundation in timing analysis fundamentals.
- Explore advanced timing performance enhancement techniques such as pipeline, retiming, and clock skew management, addressing timing analysis at both block and chip levels.
- Gain hands-on experience in Static Timing Analysis (STA) using Electronic Design Automation (EDA) tools, covering post-synthesis, place-and-route timing analysis, timing report interpretation, ECO flows execution, and achieving timing closure, ensuring practical proficiency in applying STA methodologies in real-world VLSI design projects.

Elements and Performance Criteria

Overview of VLSI STA

To be competent, the user/individual on the job must be able to:

PC1. Gain proficiency in Static Timing Analysis (STA) for VLSI design, encompassing analysis of timing in combinational and sequential circuits, resolution of timing issues like races and setup/hold violations through practical examples, and the establishment of synthesis-compatible timing constraints during circuit synthesis.

Timing performance

To be competent, the user/individual on the job must be able to:

PC2. Apply advanced techniques in timing performance enhancement, such as pipelining, retiming, and mitigation of clock skew issues, while conducting comprehensive timing analysis at block and chip levels to optimize design integrity and operational frequency across open and closed loop timing scenarios.

STA using EDA tools

To be competent, the user/individual on the job must be able to:

PC3. Utilize Electronic Design Automation (EDA) tools to apply Static Timing Analysis (STA), ensuring thorough timing analysis post synthesis and place-and-route stages to validate design timing constraints. Evaluate timing reports, manage Engineering Change Orders (ECO) flows, and conduct sign-off checks to achieve timing closure.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Develop a comprehensive understanding of timing analysis principles, encompassing races, hazards, setup and hold timing, and maximum frequency of operation for both combinational and sequential circuits.

KU2. Recognize the importance of timing constraints in the synthesis process, gaining insight into circuit synthesis methodologies with a specific focus on timing analysis

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to analyze complex timing issues in VLSI design, such as races, hazards, and setup/hold time violations, and devise effective solutions and Capacity to identify timing constraints and challenges during circuit synthesis and timing analysis, and apply analytical thinking to address them.

GS2. Proficiency in using Electronic Design Automation (EDA) tools for Static Timing Analysis (STA), including post-synthesis and place-and-route timing analysis, and interpreting timing reports.

GS3. Competence in executing Engineering Change Order (ECO) flows, conducting timing closure, and performing sign-off checks, demonstrating technical proficiency in utilizing EDA tools for VLSI design

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Overview of VLSI STA	**30**	**25**	**-**	**6**
PC1. Gain proficiency in Static Timing Analysis (STA) for VLSI design, encompassing analysis of timing in combinational and sequential circuits, resolution of timing issues like races and setup/hold violations through practical examples, and the establishment of synthesis-compatible timing constraints during circuit synthesis.	-	-	-	-
Timing performance	**30**	**25**	**-**	**7**
PC2. Apply advanced techniques in timing performance enhancement, such as pipelining, retiming, and mitigation of clock skew issues, while conducting comprehensive timing analysis at block and chip levels to optimize design integrity and operational frequency across open and closed loop timing scenarios.	-	-	-	-
STA using EDA tools	**40**	**30**	**-**	**7**
PC3. Utilize Electronic Design Automation (EDA) tools to apply Static Timing Analysis (STA), ensuring thorough timing analysis post synthesis and place-and-route stages to validate design timing constraints. Evaluate timing reports, manage Engineering Change Orders (ECO) flows, and conduct sign-off checks to achieve timing closure.	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0103
NOS Name	Static Timing Analysis of VLSI Circuits
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0104: FPGA Architecture and Programming

Description

This course offers a comprehensive study of Field-Programmable Gate Arrays (FPGAs), exploring their applications and advantages in digital design. Students delve into FPGA architecture, understanding its components like Look-Up Tables (LUTs) and flip-flops. They learn to write RTL code tailored for FPGA implementation through hands-on practice with Hardware Description Languages (HDLs) like Verilog. The curriculum covers the FPGA design flow from entry to bitstream generation, alongside design constraints. Advanced topics include FPGA design optimization techniques, verification, and debugging strategies, ensuring proficiency in FPGA-based projects.

Scope

The scope covers the following :

- The course will equip students with practical skills in Hardware Description Languages (HDLs), particularly Verilog, enabling them to proficiently write RTL code tailored for FPGA implementation. They will also gain a thorough understanding of the FPGA design flow, covering design entry, synthesis, placement, routing, and bitstream generation, alongside essential design constraints. Moreover, students will explore advanced optimization techniques like pipelining, retiming, and trade-offs between area, power, and performance to enhance FPGA designs. Additionally, they will learn FPGA verification and debugging methodologies, including simulation, testbench development, test vector creation, and effective debugging strategies, ensuring the development of robust and reliable FPGA designs.

Elements and Performance Criteria

Introduction to FPGAs

To be competent, the user/individual on the job must be able to:

PC1. Explain the fundamental concepts of FPGAs and their role in digital design.

PC2. Describe various applications of FPGAs in modern electronics

Hardware Description Languages for FPGAs

To be competent, the user/individual on the job must be able to:

PC3. Demonstrate knowledge of Verilog syntax and data types

PC4. Write effective RTL (Register Transfer Level) code for FPGA implementations.

FPGA Design Flow and Optimization Technique

To be competent, the user/individual on the job must be able to:

PC5. Implement design constraints effectively to meet specified requirements.

PC6. Apply optimization techniques such as timing analysis, pipelining, and retiming to improve FPGA performance

FPGA Verification and Debugging

To be competent, the user/individual on the job must be able to:

PC7. Apply debugging techniques to identify and resolve issues in FPGA designs

PC8. Utilize simulation tools to ensure designs meet required specifications and performance criteria.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Develop a comprehensive understanding of FPGAs, including their applications and advantages in digital design, the internal architecture featuring components like Look-Up Tables (LUTs), flip-flops, and interconnects, as well as exploring different FPGA families and their available resources, such as memory blocks and DSP slices.

KU2. Master the use of Hardware Description Languages (HDLs), particularly Verilog, for writing RTL code for FPGAs, comprehend the complete FPGA design flow from design entry to bitstream generation, including the role of design constraints,and learn optimization techniques like pipelining, retiming, and balancing trade-offs between area, power, and performance to improve FPGA design efficiency

KU3. Grasp the importance of FPGA simulation and verification by learning to write effective test benches and test vectors, and acquire debugging skills using tools and techniques such as logic analyzers and on-chip debugging tools to ensure correct and optimized functionality of FPGA designs.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Gain comprehensive expertise in FPGA architecture, Verilog proficiency, and the complete FPGA design process, covering design entry to bitstream generation with the application of design constraints. Develop skills in analyzing and optimizing designs using techniques like pipelining and retiming, while balancing trade-offs between area, power, and performance.

GS2. Develop proficiency in FPGA design validation through simulation, leveraging effective test benches and test vectors. Master debugging tools and techniques to ensure the accuracy and optimization of FPGA functionality. Improve problem-solving skills and attention to detail by identifying and resolving design issues throughout the development process

GS3. Develop proficiency in managing FPGA projects from inception to implementation by applying systematic design approaches. Adapt to diverse FPGA families and resources, staying abreast of advancements in technology and tools. Enhance collaboration and communication skills through effective teamwork and clear articulation of design decisions and optimization strategies.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to FPGAs	**25**	**20**	**-**	**5**
PC1. Explain the fundamental concepts of FPGAs and their role in digital design.	-	-	-	-
PC2. Describe various applications of FPGAs in modern electronics	-	-	-	-
Hardware Description Languages for FPGAs	**25**	**20**	**-**	**5**
PC3. Demonstrate knowledge of Verilog syntax and data types	-	-	-	-
PC4. Write effective RTL (Register Transfer Level) code for FPGA implementations.	-	-	-	-
FPGA Design Flow and Optimization Technique	**25**	**20**	**-**	**5**
PC5. Implement design constraints effectively to meet specified requirements.	-	-	-	-
PC6. Apply optimization techniques such as timing analysis, pipelining, and retiming to improve FPGA performance	-	-	-	-
FPGA Verification and Debugging	**25**	**20**	**-**	**5**
PC7. Apply debugging techniques to identify and resolve issues in FPGA designs	-	-	-	-
PC8. Utilize simulation tools to ensure designs meet required specifications and performance criteria.	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0104
NOS Name	FPGA Architecture and Programming
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0105: VLSI Verification fundamentals

Description

This NOS is designed to provide a comprehensive understanding of the essential principles, methodologies, and tools involved in the verification of Very Large Scale Integration (VLSI) circuits. This educational program begins by establishing a solid foundation in digital design concepts, ensuring that participants possess a strong grasp of the fundamental building blocks of VLSI systems. The curriculum then progresses to explore the intricacies of verification methodologies, emphasizing the significance of thorough testing and validation in the semiconductor design process.

Scope

The scope covers the following :

- This includes understanding lexical conventions such as port types, data types, operators, and system tasks. It covers design abstractions like logic primitives, gate-level modeling, data flow modeling, and behavioral modeling in Verilog HDL.
- Students will learn hierarchical modeling methodologies and port connection rules. Concepts include module instantiation by port name or order, creating test benches, and managing design abstractions files essential for structured verification.
- The syllabus emphasizes verification architecture and flow, encompassing design verification using Verilog HDL and various types of test benches. It covers coverage-driven verification techniques, writing effective test cases, and employing scripting languages for test automation, preparing students for rigorous verification practices in digital design.

Elements and Performance Criteria

Basics of Verilog HDL for verification

To be competent, the user/individual on the job must be able to:

PC1. Understanding of Verilog HDL lexical conventions including port types, data types, operators, numbers, vectors, arrays, compiler directives, and system tasks

PC2. Ability to implement Verilog HDL design abstractions such as logic primitives, gate-level modeling, data flow modeling, and behavioral modeling

Hierarchal Modeling and test benching for verification

To be competent, the user/individual on the job must be able to:

PC3. Competency in applying design methodologies and adhering to port connection rules in hierarchical modeling

PC4. Capability to develop and utilize test benches effectively for verifying Verilog HDL designs

Introduction to Verification, Verification Types, Verification Architecture and Flow

To be competent, the user/individual on the job must be able to:

PC5. Understanding of the verification process, including verification architecture and flow in digital design projects

PC6. Knowledge of various verification types such as test automation, assertions, and coverage-driven verification

Test Automation and Scripting

To be competent, the user/individual on the job must be able to:

PC7. Proficiency in using scripting languages for test automation in Verilog HDL verification

PC8. Capability to develop scripts to automate test cases, improve verification efficiency, and manage verification components

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Comprehensive understanding of Verilog HDL basics, encompassing lexical conventions, design abstractions, and modeling techniques like gate-level and behavioral modeling

KU2. Proficiency in hierarchical modeling methodologies, including module instantiation rules and the utilization of test benches for structured design verification

KU3. Knowledge and application of advanced verification techniques, encompassing verification architecture, test automation scripting, and coverage-driven verification to ensure robust design validation

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to write and understand Verilog HDL code, including lexical conventions, design abstractions, and behavioral modeling techniques

GS2. Capability to apply hierarchical modeling methodologies for structured design, manage port connections, and instantiate modules using different methods

GS3. Skills in verification architecture, flow management, and employing various verification techniques such as test automation, assertions, and coverage-driven verification

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Basics of Verilog HDL for verification	**25**	**20**	**-**	**5**
PC1. Understanding of Verilog HDL lexical conventions including port types, data types, operators, numbers, vectors, arrays, compiler directives, and system tasks	-	-	-	-
PC2. Ability to implement Verilog HDL design abstractions such as logic primitives, gate-level modeling, data flow modeling, and behavioral modeling	-	-	-	-
Hierarchal Modeling and test benching for verification	**25**	**20**	**-**	**5**
PC3. Competency in applying design methodologies and adhering to port connection rules in hierarchical modeling	-	-	-	-
PC4. Capability to develop and utilize test benches effectively for verifying Verilog HDL designs	-	-	-	-
Introduction to Verification, Verification Types, Verification Architecture and Flow	**25**	**20**	**-**	**5**
PC5. Understanding of the verification process, including verification architecture and flow in digital design projects	-	-	-	-
PC6. Knowledge of various verification types such as test automation, assertions, and coverage-driven verification	-	-	-	-
Test Automation and Scripting	**25**	**20**	**-**	**5**
PC7. Proficiency in using scripting languages for test automation in Verilog HDL verification	-	-	-	-
PC8. Capability to develop scripts to automate test cases, improve verification efficiency, and manage verification components	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0105
NOS Name	VLSI Verification fundamentals
Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	VLSI Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0106: ASIC Verification using System Verilog and UVM

Description

This NOS covers comprehensive training in integrated circuit and processor verification, spanning Verilog HDL and System Verilog for RTL verification, advanced functional verification techniques, scripting languages for automation, processor architecture including RISC-V and DIR-V microprocessors, SoC verification methodologies, and FPGA-based emulation for SoC validation and performance evaluation.

Scope

The scope covers the following :

- The NOS covers Verilog HDL and System Verilog for integrated circuit verification, emphasizing RTL verification, test bench creation, and automation with scripting languages. It includes detailed exploration of processor architecture, encompassing RISC-V and DIR-V microprocessors, focusing on simulation, verification planning, and functional coverage assessment. The module also delves into SoC verification methodologies, including AXI bus architecture, UVM-based verification environments, and FPGA emulation for post-silicon validation and performance evaluation.

Elements and Performance Criteria

Integrated Circuit verification

To be competent, the user/individual on the job must be able to:

PC1. Proficient in Verilog HDL for RTL verification, encompassing the creation of diverse test benches, formulation of comprehensive test cases, and implementation of streamlined verification architectures

PC2. Proficiency in System Verilog for advanced verification techniques, utilizing simulators, and effectively applying coverage metrics for comprehensive verification

PC3. Competence in scripting languages like Linux shell scripting, Perl, and Python for automation in verification processes

Processor Architecture and Processor verification, SoC Verification

To be competent, the user/individual on the job must be able to:

PC4. Proficiency in processor architecture and verification, including understanding ISA concepts, memory hierarchy, and advanced architectures like superscalar processors, accelerators, and vector processors. Extensive knowledge of DIR-V microprocessors, focusing on microcontroller-class features, addressing schemes, memory mapping, and arbitration protocols

PC5. Expertise in system-on-chip (SoC) verification, particularly with DIR-V processor-based SoCs. In-depth understanding of SoC architectures, including peripheral components, memory structures, and basic building blocks. Mastery of verification methodologies such as assertion-based verification (ABV), coverage-driven verification (CDV), and constraint random verification (CRV). Skilled in developing verification IPs and environments

PC6. Practical proficiency in Universal Verification Methodology (UVM), enabling the construction of effective SoC verification environments. Experience in writing advanced test cases and executing verification projects involving DIR-V and Swadeshi processor-based SoCs

Methodology based SoC Verification, SoC Emulation and post silicon Validation

To be competent, the user/individual on the job must be able to:

PC7. Proficient in Universal Verification Methodology (UVM), capable of constructing robust SoC verification environments

PC8. Skilled in writing complex test cases to ensure thorough verification of DIR-V and Swadeshi processor-based SoCs

PC9. Experienced in executing verification projects, applying UVM methodologies effectively to achieve reliable SoC validation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Proficiency in Verilog HDL for RTL verification, encompassing test bench creation, test case development, verification architecture, automation, assertions, and coverage metrics

KU2. Expertise in processor architecture, including ISA fundamentals, memory hierarchy, DIR-V microprocessor features like address mapping and arbitration, and simulation for verification, with emphasis on functional coverage

KU3. Comprehensive understanding and practical application of SoC verification methodologies, including UVM-based environments, IP verification, test case automation, and validation using FPGA emulation for DIR-V processor-based designs

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Proficient in Verilog HDL and System Verilog, mastering RTL verification through test bench creation, test case development, and setting up verification architectures, alongside expertise in C and object-oriented programming essential

GS2. Proficient application of Universal Verification Methodology (UVM) to build robust SoC verification environments, incorporating assertion-based verification (ABV), coverage-driven verification (CDV), and constraint random verification (CRV) for comprehensive verification coverage.

GS3. Comprehensive expertise in processor architecture, encompassing instruction set architecture (ISA), memory hierarchy, and advanced features like superscalar and vector processors, alongside specialized knowledge of DIR-V microprocessors, addressing mechanisms, bus architecture, and memory mapping crucial for robust processor simulation and verification.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Integrated Circuit verification	**30**	**25**	**-**	**6**
PC1. Proficient in Verilog HDL for RTL verification, encompassing the creation of diverse test benches, formulation of comprehensive test cases, and implementation of streamlined verification architectures	-	-	-	-
PC2. Proficiency in System Verilog for advanced verification techniques, utilizing simulators, and effectively applying coverage metrics for comprehensive verification	-	-	-	-
PC3. Competence in scripting languages like Linux shell scripting, Perl, and Python for automation in verification processes	-	-	-	-
Processor Architecture and Processor verification, SoC Verification	**30**	**25**	**-**	**7**
PC4. Proficiency in processor architecture and verification, including understanding ISA concepts, memory hierarchy, and advanced architectures like superscalar processors, accelerators, and vector processors. Extensive knowledge of DIR-V microprocessors, focusing on microcontroller-class features, addressing schemes, memory mapping, and arbitration protocols	-	-	-	-
PC5. Expertise in system-on-chip (SoC) verification, particularly with DIR-V processor-based SoCs. In-depth understanding of SoC architectures, including peripheral components, memory structures, and basic building blocks. Mastery of verification methodologies such as assertion-based verification (ABV), coverage-driven verification (CDV), and constraint random verification (CRV). Skilled in developing verification IPs and environments	-	-	-	-
PC6. Practical proficiency in Universal Verification Methodology (UVM), enabling the construction of effective SoC verification environments. Experience in writing advanced test cases and executing verification projects involving DIR-V and Swadeshi processor-based SoCs	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Methodology based SoC Verification, SoC Emulation and post silicon Validation	**40**	**30**	**-**	**7**
PC7. Proficient in Universal Verification Methodology (UVM), capable of constructing robust SoC verification environments	-	-	-	-
PC8. Skilled in writing complex test cases to ensure thorough verification of DIR-V and Swadeshi processor-based SoCs	-	-	-	-
PC9. Experienced in executing verification projects, applying UVM methodologies effectively to achieve reliable SoC validation	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0106
NOS Name	ASIC Verification using System Verilog and UVM
Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	VLSI Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0107: VLSI Circuits Design for testability

Description

This NOS provides a comprehensive exploration of Design for Test (DFT) in VLSI circuits, covering fundamental concepts such as the importance and implementation of DFT principles for detecting manufacturing faults and design defects essential for circuit reliability. It includes detailed studies on Built-In Self-Test (BIST) methodologies like Memory BIST (MBIST) and Logic BIST (LBIST), alongside Automatic Test Pattern Generation (ATPG) techniques for efficient fault detection.

Scope

The scope covers the following :

- This NOS Covers essential concepts such as the role of DFT in identifying manufacturing faults and design defects, emphasizing the necessity of testing in ensuring circuit reliability, delving into ATPG methodologies including BIST (Built-In Self-Test), MBIST (Memory BIST), and LBIST (Logic BIST), alongside Boundary Scan techniques for comprehensive testing coverage. Exploring DFT compiler workflows and scan insertion flows for efficient test pattern generation. JTAG for external testing through Registers and the Test Access Port (TAP), and IJTAG for internal testing with features like Segment Insertion Bit (SIB), Instrument Connectivity Language (ICL), and Procedural Description Language (PDL), crucial for intricate test strategy implementations in VLSI circuits.

Elements and Performance Criteria

Overview of DFT for VLSI

To be competent, the user/individual on the job must be able to:

PC1. Understand the fundamentals of Design for Test (DFT), including the identification of manufacturing faults and design defects

PC2. Demonstrate proficiency in Scan ATPG (Automatic Test Pattern Generation) and fault simulation techniques

PC3. Apply scan design rules and conduct scan tests focusing on transition delays

ATPG and BIST

To be competent, the user/individual on the job must be able to:

PC4. Implement Built-In Self-Test (BIST) techniques, including MBIST (Memory BIST) and LBIST (Logic BIST).

PC5. Utilize Boundary Scan techniques for testing and debugging using Boundary Scan Cells and Instructions

PC6. Execute flows supported by DFT compilers for scan insertion and manage DFT compiler flows and commands proficiently

JTAG and IJTAG

To be competent, the user/individual on the job must be able to:

PC7. Describe the JTAG standard and its components, including Registers and the Test Access Port (TAP).

PC8. Implement IJTAG for internal component testing, utilizing Segment Insertion Bit (SIB) functionalities

PC9. Script and automate test procedures using Procedural Description Language (PDL) in IJTAG environments for efficient testing and validation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Gain expertise in Design for Test (DFT), essential for identifying manufacturing faults and design defects in VLSI circuits, employing methods like Scan ATPG, Fault Simulation, and scan design rules to ensure thorough testing, including targeting transition delays.

KU2. Develop comprehensive skills in Built-In Self-Test (BIST), covering MBIST and LBIST methodologies for efficient memory and logic testing. Additionally, master Boundary Scan techniques integrated into DFT workflows via DFT compilers, alongside proficiency in scan insertion flows and command utilization

KU3. Become proficient in JTAG fundamentals, emphasizing Registers and the Test Access Port (TAP) for external testing. Explore Internal JTAG (IJTAG) capabilities for internal component testing in VLSI circuits. Master the use of Instrument Connectivity Language (ICL) and Procedural Description Language (PDL) for configuring test instruments and automating test procedures in IJTAG environments.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Understanding fundamental principles of Design for Test (DFT), including identifying manufacturing faults, design defects, and the necessity of testing for ensuring VLSI circuit reliability and performance

GS2. Proficiency in Scan-based Automatic Test Pattern Generation (ATPG) and Built-In Self-Test (BIST) methodologies (MBIST, LBIST), essential for comprehensive memory and logic testing

GS3. Ability to integrate Boundary Scan techniques into DFT workflows using DFT compilers, along with mastery of scan insertion flows and command usage within DFT compiler environments.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Overview of DFT for VLSI	**30**	**25**	**-**	**6**
PC1. Understand the fundamentals of Design for Test (DFT), including the identification of manufacturing faults and design defects	-	-	-	-
PC2. Demonstrate proficiency in Scan ATPG (Automatic Test Pattern Generation) and fault simulation techniques	-	-	-	-
PC3. Apply scan design rules and conduct scan tests focusing on transition delays	-	-	-	-
ATPG and BIST	**30**	**25**	**-**	**7**
PC4. Implement Built-In Self-Test (BIST) techniques, including MBIST (Memory BIST) and LBIST (Logic BIST).	-	-	-	-
PC5. Utilize Boundary Scan techniques for testing and debugging using Boundary Scan Cells and Instructions	-	-	-	-
PC6. Execute flows supported by DFT compilers for scan insertion and manage DFT compiler flows and commands proficiently	-	-	-	-
JTAG and IJTAG	**40**	**30**	**-**	**7**
PC7. Describe the JTAG standard and its components, including Registers and the Test Access Port (TAP).	-	-	-	-
PC8. Implement IJTAG for internal component testing, utilizing Segment Insertion Bit (SIB) functionalities	-	-	-	-
PC9. Script and automate test procedures using Procedural Description Language (PDL) in IJTAG environments for efficient testing and validation.	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0107
NOS Name	VLSI Circuits Design for testability
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0108: VLSI Physical Design and Verification

Description

This NOS provides an in-depth introduction to VLSI Physical Design, covering key concepts such as the physical design flow, CMOS fabrication processes, and essential design steps from RTL to GDSII. Students will learn about placement, routing, clock tree synthesis, power planning, and verification techniques like functional, timing, and signal integrity verification.

Scope

The scope covers the following :

- The scope of this course encompasses the entire VLSI physical design process, from RTL coding to final chip layout (GDSII). It covers key topics such as CMOS fabrication, design flow, placement, routing, power planning, and clock tree synthesis. Additionally, students will gain hands-on experience with industry-standard EDA tools for layout, simulation, and verification, and analyze real-world design challenges, preparing them for professional roles in VLSI design and verification.

Elements and Performance Criteria

Introduction to Physical Design ,VLSI Physical Design Flow

To be competent, the user/individual on the job must be able to:

PC1. Demonstrated understanding of VLSI physical design concepts, including the physical design flow from RTL (Register Transfer Level) to GDSII (Graphic Data System II).

PC2. Ability to describe the industry-standard flow for VLSI physical design, encompassing fabrication processes such as CMOS technology and photolithography

Verification Techniques

To be competent, the user/individual on the job must be able to:

PC3. Proficiency in functional verification methodologies, utilizing simulation tools and testbenches to ensure design correctness and coverage analysis

PC4. Competency in timing verification using Static Timing Analysis (STA) to identify and mitigate timing violations and uncertainties

Design Tools and EDA

To be competent, the user/individual on the job must be able to:

PC5. Practical knowledge of Electronic Design Automation (EDA) tools for layout editing, including place and route tools and the use of cell libraries

PC6. Ability to use verification tools like Static Timing Analysis (STA) tools and Layout vs. Schematic (LVS) checkers to ensure design accuracy and compliance with specifications

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Gain a foundational understanding of VLSI physical design, encompassing the industry-standard flow from RTL to GDSII. Explore RTL design and synthesis techniques using Verilog/VHDL, along with strategies for placement, routing, clock tree synthesis, and optimization of power grids

KU2. Gain expertise in essential verification methodologies crucial for VLSI design, encompassing functional verification through simulation and comprehensive coverage analysis using testbenches.

KU3. Acquire proficiency in Electronic Design Automation (EDA) tools used in VLSI design, covering layout editors, place and route tools, and cell libraries.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Proficiency in utilizing Electronic Design Automation (EDA) tools for layout editing, including place and route functionalities and management of cell libraries.

GS2. Competence in employing simulation tools such as Verilog/VHDL simulators and waveform viewers to validate functional correctness and performance metrics

GS3. Ability to conduct various verification techniques, including Static Timing Analysis (STA) and Layout vs. Schematic (LVS) checking, to ensure design integrity and adherence to specifications

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Physical Design ,VLSI Physical Design Flow	**30**	**25**	**-**	**6**
PC1. Demonstrated understanding of VLSI physical design concepts, including the physical design flow from RTL (Register Transfer Level) to GDSII (Graphic Data System II).	-	-	-	-
PC2. Ability to describe the industry-standard flow for VLSI physical design, encompassing fabrication processes such as CMOS technology and photolithography	-	-	-	-
Verification Techniques	**30**	**25**	**-**	**7**
PC3. Proficiency in functional verification methodologies, utilizing simulation tools and testbenches to ensure design correctness and coverage analysis	-	-	-	-
PC4. Competency in timing verification using Static Timing Analysis (STA) to identify and mitigate timing violations and uncertainties	-	-	-	-
Design Tools and EDA	**40**	**30**	**-**	**7**
PC5. Practical knowledge of Electronic Design Automation (EDA) tools for layout editing, including place and route tools and the use of cell libraries	-	-	-	-
PC6. Ability to use verification tools like Static Timing Analysis (STA) tools and Layout vs. Schematic (LVS) checkers to ensure design accuracy and compliance with specifications	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0108
NOS Name	VLSI Physical Design and Verification
Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	VLSI Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0109: Accelerator design using HLS programming

Description

This course covers advanced techniques in High-Level Synthesis (HLS) for designing combination and sequential circuits using C/C++. Students will explore the HLS design flow, ports, and propagation delay computation, as well as develop test benches. Topics include IP-centric design flow for sequential circuits, state machines, functional pipelining, and interface synthesis. Additionally, function acceleration on FPGA will be explored through OpenCL concepts, including kernel execution, loop optimization, and memory dependencies.

Scope

The scope covers the following :

- The scope of this course encompasses the design and implementation of both combinational and sequential circuits using High-Level Synthesis (HLS) and C/C++. It provides in-depth knowledge of HLS design flow, test bench development, and optimization techniques for FPGA implementation. Students will also learn about function acceleration using OpenCL, focusing on loop optimization, memory dependencies, and real-time performance improvements, preparing them for advanced FPGA design roles.

Elements and Performance Criteria

Combination Circuits and Test Benches using C/C++

To be competent, the user/individual on the job must be able to:

PC1. Able to design combinational logic circuits with C/C++ language using the HLS approach

PC2. Learn to work with Xilinx Vitis-HLS and Vivado suite Toolsets

PC3. Able to generate RTL hardware IPs using Vitis-HLS

Sequential Circuits Design using HLS

To be competent, the user/individual on the job must be able to:

PC4. Able to use HLS concepts for designing sequential logic circuits

PC5. Learn to Write C-testbench in HLS

PC6. Able to optimize the design using the Interfaces

Function Acceleration on FPGA

To be competent, the user/individual on the job must be able to:

PC7. Understand the concepts of FPGA-Based embedded systems

PC8. Able to accelerate compute-intensive algorithms on Zynq platforms Using C/C++/OpenCL

PC9. Able to run the accelerated applications on FPGAs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Expertise in developing combination circuits using C/C++, including creating robust test benches for functional validation, managing data types and conditional statements, utilizing bit precision libraries for accuracy, and optimizing designs through combinatorial loop unrolling and resource management techniques.

KU2. Proficiency in HLS for sequential circuit design, covering single-cycle methodologies and IP-centric flows for parallel-to-serial and serial-to-parallel conversions, alongside implementing state machines, timers, counters, debounce circuits, clock generators for precise timing, and functional pipelining with interface synthesis to optimize performance and integration of complex designs

KU3. Proficiency in OpenCL concepts for FPGA acceleration, emphasizing kernel execution models, efficient data transfer strategies, loop latency optimization, array partitioning, loop merging, pipeline implementation, and resolving data and memory dependencies to enhance FPGA computational efficiency in function acceleration applications

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Proficiency in programming combination circuits and developing robust test benches using C/C++ for functional verification.

GS2. Ability to utilize HLS for sequential circuit design, including single-cycle methodologies and IP-centric flows for parallel and serial data processing

GS3. Competence in optimizing designs through techniques such as combinatorial loop unrolling, managing overloading, and addressing resource constraints

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Combination Circuits and Test Benches using C/C++	**30**	**25**	**-**	**6**
PC1. Able to design combinational logic circuits with C/C++ language using the HLS approach	-	-	-	-
PC2. Learn to work with Xilinx Vitis-HLS and Vivado suite Toolsets	-	-	-	-
PC3. Able to generate RTL hardware IPs using Vitis-HLS	-	-	-	-
Sequential Circuits Design using HLS	**30**	**25**	**-**	**7**
PC4. Able to use HLS concepts for designing sequential logic circuits	-	-	-	-
PC5. Learn to Write C-testbench in HLS	-	-	-	-
PC6. Able to optimize the design using the Interfaces	-	-	-	-
Function Acceleration on FPGA	**40**	**30**	**-**	**7**
PC7. Understand the concepts of FPGA-Based embedded systems	-	-	-	-
PC8. Able to accelerate compute-intensive algorithms on Zynq platforms Using C/C++/OpenCL	-	-	-	-
PC9. Able to run the accelerated applications on FPGAs	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0109
NOS Name	Accelerator design using HLS programming
Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	VLSI Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0110: SOC Design and Verification

Description

This course provides an in-depth understanding of advanced System-on-Chip (SoC) design and verification methodologies. It covers multi-core SoCs, hardware accelerators, and UVM-based SoC verification environments, including hands-on case studies like DIR-V processor verification. Additionally, the course explores SoC emulation, FPGA-based verification, post-silicon validation, and high-level synthesis for AI/ML hardware accelerators. Students will engage in real-world projects, including Linux porting, benchmarking, and performance evaluation, preparing them for modern SoC development and testing challenges.

Scope

The scope covers the following :

- The scope of this course encompasses advanced SoC design, verification, and validation methodologies, focusing on multi-core SoCs, hardware accelerators, and Universal Verification Methodology (UVM). It also covers FPGA-based emulation, post-silicon validation, and high-level synthesis for AI/ML hardware. Students will gain hands-on experience in SoC testing, performance evaluation, Linux porting, and application development, preparing them for industry roles in modern SoC development and verification

Elements and Performance Criteria

Advanced SoCs and methodology based SoC Verification

To be competent, the user/individual on the job must be able to:

PC1. Gain comprehensive knowledge of Multi-Core SoCs, including their architectural nuances and effective integration strategies with hardware accelerators and peripheral interfaces

PC2. Utilize Universal Verification Methodology (UVM) components proficiently to construct robust test benches and verification environments tailored to SoC designs

PC3. Demonstrate proficiency in conducting IP verification using UVM through practical application in a specified case study scenario

PC4. Develop and deploy a comprehensive UVM-based SoC verification environment, emphasizing the creation and automation of test cases to ensure thorough verification coverage

SoC Emulation and Post silicon verification and validation

To be competent, the user/individual on the job must be able to:

PC5. Gain expertise in the design flow of AMD/Xilinx FPGA architectures, focusing on effective emulation techniques for SoC applications

PC6. Apply advanced logic synthesis and timing concepts to optimize FPGA designs for System-on-Chip applications

PC7. Utilize embedded logic analyzers proficiently for debugging and verifying FPGA-based SoC designs, ensuring robust functionality

PC8. Demonstrate expertise in porting and emulating DIR-V processor-based SoCs on FPGA platforms, including the development and rigorous testing of embedded ‘C’ programs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Gain in-depth knowledge of Multi-Core SoCs, encompassing their architectures and effective integration with hardware accelerators and peripheral interfaces

KU2. Gain mastery in UVM components and methodologies to construct robust test benches, perform IP verification, and create comprehensive SoC verification environments.

KU3. Gain mastery in popular AMD/Xilinx FPGA architectures and their design flow to proficiently emulate SoCs. Apply advanced logic synthesis and timing concepts to optimize FPGA designs for diverse applications, ensuring effective emulation. Utilize embedded logic analyzers competently for thorough debugging and verification during FPGA emulation processes.

KU4. Develop proficiency in porting and emulating DIR-V processor-based SoCs on FPGA platforms, including the creation and validation of embedded -C programs.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Gain expertise in Universal Verification Methodology (UVM) components for building robust test benches, conducting IP verification, and automating test cases in SoC environments

GS2. Understand the architecture and integration of Multi-Core SoCs with hardware accelerators and peripheral interfaces

GS3. Gain mastery in FPGA design flow, logic synthesis, and timing optimization tailored for SoC applications using popular AMD/Xilinx architectures. Develop expertise in porting and emulating DIR-V processor-based SoCs on FPGA platforms, encompassing embedded software development and Linux application execution.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Advanced SoCs and methodology based SoC Verification	**50**	**40**	**-**	**10**
PC1. Gain comprehensive knowledge of Multi-Core SoCs, including their architectural nuances and effective integration strategies with hardware accelerators and peripheral interfaces	-	-	-	-
PC2. Utilize Universal Verification Methodology (UVM) components proficiently to construct robust test benches and verification environments tailored to SoC designs	-	-	-	-
PC3. Demonstrate proficiency in conducting IP verification using UVM through practical application in a specified case study scenario	-	-	-	-
PC4. Develop and deploy a comprehensive UVM-based SoC verification environment, emphasizing the creation and automation of test cases to ensure thorough verification coverage	-	-	-	-
SoC Emulation and Post silicon verification and validation	**50**	**40**	**-**	**10**
PC5. Gain expertise in the design flow of AMD/Xilinx FPGA architectures, focusing on effective emulation techniques for SoC applications	-	-	-	-
PC6. Apply advanced logic synthesis and timing concepts to optimize FPGA designs for System-on-Chip applications	-	-	-	-
PC7. Utilize embedded logic analyzers proficiently for debugging and verifying FPGA-based SoC designs, ensuring robust functionality	-	-	-	-
PC8. Demonstrate expertise in porting and emulating DIR-V processor-based SoCs on FPGA platforms, including the development and rigorous testing of embedded ‘C’ programs	-	-	-	-
NOS Total	**100**	**80**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0110
NOS Name	SOC Design and Verification
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

NIE/ELE/N0111: Implementation of Advanced VLSI Circuit Design and Verification using Verilog, FPGA, and System-on-Chip (SoC) Architectures

Description

This NOS focuses on designing and verifying advanced VLSI circuits using Verilog, FPGA, and System-on-Chip (SoC) architectures. Students will learn to develop digital circuits, perform static timing analysis, and optimize circuit layouts. The course covers FPGA programming, synthesis, and physical design flows, providing hands-on experience with industry-standard tools. It also introduces advanced verification techniques using System Verilog and UVM, ensuring reliable and efficient SoC designs.

Scope

The scope covers the following :

- The scope of this project includes developing proficiency in VLSI circuit design using Verilog for RTL coding, focusing on FPGA and System-on-Chip (SoC) architectures. It covers the full design flow from RTL to GDSII, including synthesis, static timing analysis, and physical design optimization. Students will also explore advanced verification methodologies using System Verilog and UVM, ensuring robust validation of SoC designs. The project provides practical experience in FPGA-based implementations, circuit optimization, and testing strategies. By completing the project, students will gain a comprehensive understanding of modern VLSI design and verification processes.

Elements and Performance Criteria

Implementation of Advanced VLSI Circuit Design and Verification using Verilog, FPGA, and System-on-Chip (SoC) Architectures

To be competent, the user/individual on the job must be able to:

PC1. Students should demonstrate the ability to accurately implement specified digital circuits in Verilog, ensuring that the functionality matches the intended design.

PC2. Students should effectively use simulation tools (e.g., ModelSim) to run simulations of their Verilog designs and produce correct waveforms that reflect the expected behavior.

PC3. Students should be able to create comprehensive testbenches that thoroughly test the functionality of their designs, identifying and fixing any issues or bugs that arise.

PC4. Students should successfully synthesize and configure their designs on the FPGA, demonstrating understanding of the design flow from synthesis to programming.

PC5. Students should optimize their designs for performance metrics such as speed and resource utilization, analyzing the impact of different design choices on FPGA performance.

PC6. Students should demonstrate the ability to validate their FPGA implementations through physical testing, ensuring that the hardware behaves as expected in real-time scenarios.

PC7. Students should effectively integrate various components (e.g., microcontrollers, memory, peripherals) into a cohesive SoC design that meets specified functional requirements

PC8. Students should apply appropriate verification methodologies (such as simulation and formal verification) to ensure the reliability and correctness of their SoC designs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the basic syntax, data types, and operators in Verilog, enabling them to write effective code for digital circuit design.

KU2. Comprehend the simulation process, including waveform analysis and debugging, to ensure their designs function correctly under different conditions.

KU3. Familiar with the architecture of FPGAs, including the roles of CLBs, I/O blocks, and interconnects, which are crucial for effective design implementation.

KU4. Understand the various components of an SoC, including processors, memory, and peripherals, and how they interact within a system.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Students should develop the ability to identify design challenges and implement effective solutions when coding in Verilog and debugging simulations.

GS2. Students should engage in collaborative work, fostering communication and teamwork skills while working on FPGA design projects, simulating real-world engineering environments.

GS3. Students should develop critical evaluation skills to assess the effectiveness and reliability of their SoC designs, considering factors such as performance, scalability, and testability.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Implementation of Advanced VLSI Circuit Design and Verification using Verilog, FPGA, and System-on-Chip (SoC) Architectures	**-**	**-**	**80**	**20**
PC1. Students should demonstrate the ability to accurately implement specified digital circuits in Verilog, ensuring that the functionality matches the intended design.	-	-	-	-
PC2. Students should effectively use simulation tools (e.g., ModelSim) to run simulations of their Verilog designs and produce correct waveforms that reflect the expected behavior.	-	-	-	-
PC3. Students should be able to create comprehensive testbenches that thoroughly test the functionality of their designs, identifying and fixing any issues or bugs that arise.	-	-	-	-
PC4. Students should successfully synthesize and configure their designs on the FPGA, demonstrating understanding of the design flow from synthesis to programming.	-	-	-	-
PC5. Students should optimize their designs for performance metrics such as speed and resource utilization, analyzing the impact of different design choices on FPGA performance.	-	-	-	-
PC6. Students should demonstrate the ability to validate their FPGA implementations through physical testing, ensuring that the hardware behaves as expected in real-time scenarios.	-	-	-	-
PC7. Students should effectively integrate various components (e.g., microcontrollers, memory, peripherals) into a cohesive SoC design that meets specified functional requirements	-	-	-	-
PC8. Students should apply appropriate verification methodologies (such as simulation and formal verification) to ensure the reliability and correctness of their SoC designs	-	-	-	-
NOS Total	**-**	**-**	**80**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0111
NOS Name	Implementation of Advanced VLSI Circuit Design and Verification using Verilog, FPGA, and System-on-Chip (SoC) Architectures
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the current job market requirement and future of work

PC2. identify and explore learning and employability relevant portals

PC3. research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC5. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC6. recognize the significance of 21st Century Skills for employment

PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

PC8. adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC9. use basic English for everyday conversation in different contexts, in person and over the telephone

PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC11. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC12. identify career goals based on the skills, interests, knowledge, and personal attributes

PC13. prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings

PC15. use active listening techniques for effective communication

PC16. communicate in writing using appropriate style and format based on formal or informal requirements

PC17. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC18. communicate and behave appropriately with all genders and PwD

PC19. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.

PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook

PC22. identify common components of salary and compute income, expenses, taxes, investments etc

PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC24. operate digital devices and use their features and applications securely and safely

PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.

PC26. display responsible online behaviour while using various social media platforms

PC27. create a personal email account, send and process received messages as per requirement

PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications

PC29. utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC33. identify different types of customers and ways to communicate with them

PC34. identify and respond to customer requests and needs in a professional manner

PC35. use appropriate tools to collect customer feedback

PC36. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC37. create a professional Curriculum vitae (Résumé)

PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC39. apply to identified job openings using offline /online methods as per requirement

PC40. answer questions politely, with clarity and confidence, during recruitment and selection

PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. components of salary and how to compute income and expenditure
KU12. importance of maintaining safety and security in offline and online financial transactions
KU13. different legal rights and laws
KU14. different types of digital devices and the procedure to operate them safely and securely
KU15. how to create and operate an e- mail account
KU16. use applications such as word processors, spreadsheets etc.
KU17. how to identify business opportunities
KU18. types and needs of customers
KU19. how to apply for a job and prepare for an interview
KU20. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence in English and other languages
GS2. communicate effectively using appropriate language in formal and informal settings
GS3. behave politely and appropriately with all to maintain effective work relationship
GS4. how to work in a virtual mode, using various technological platforms
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
Basic English Skills	**3**	**4**	**-**	**-**
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**5**	**-**	**-**
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria. The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory exams are set by the Examination wing (assessor) of NIELIT HQS.

2. The assessor assigns the roll number.

3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds

and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0101.VLSI Fundamentals	100	80	0	20	200	9
NIE/ELE/N0102.Verilog RTL coding for Synthesis	100	80	0	20	200	9
NIE/ELE/N0103.Static Timing Analysis of VLSI Circuits	100	80	0	20	200	9

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0104.FPGA Architecture and Programming	100	80	0	20	200	9
NIE/ELE/N0105.VLSI Verification fundamentals	100	80	0	20	200	9
NIE/ELE/N0106.ASIC Verification using System Verilog and UVM	100	80	0	20	200	9
NIE/ELE/N0107.VLSI Circuits Design for testability	100	80	0	20	200	9
NIE/ELE/N0108.VLSI Physical Design and Verification	100	80	0	20	200	9
NIE/ELE/N0109.Accelerator design using HLS programming	100	80	0	20	200	9
NIE/ELE/N0110.SOC Design and Verification	100	80	0	20	200	9
NIE/ELE/N0111.Implementation of Advanced VLSI Circuit Design and Verification using Verilog, FPGA, and System-on-Chip (SoC) Architectures	0	0	80	20	100	8
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	2
Total	**1020**	**830**	**80**	**220**	**2150**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
NCO	National Classification of Occupations
ISCO	International Standard Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard
NQR	National Qualification Register
NSQF	National Skills Qualification Framework
OJT	On the Job Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf